

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8497**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2023601005
Course Section:	1				

L_29_529_986_1_1505_38335



Name of the candidate. **PAARTH SUNIL DALVI**

Candidate's UID number: **2 0 2 3 6 0 1 0 0 5**

Examination class room number: **4 0 1**

Invigilator's signature with date: **4/10/23**

(To be filled in by the candidate)

EXAMINATION: **BE. ~~B~~TECH MSE**
(For example FY MCA/FY BTECH)

SEMESTER : **7**

PROGRAM: **CSE-AIML**

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: **08/10/2025 WEDNE SDAY**

COURSE NAME: **DEEP LEARNING**

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Sr. No. 8497



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1. b)



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Q2.6

1. Auto encoders are the neural network used in unsupervised learning which learns to reconstruct the input output that resembles the input from a lower dimensional latent space vector.

2. Consists of three layers.

• Encoder :-

- Converts the input into a latent space representation.

• Bottleneck :-

- Captures important features from the lower dimension data.

• Decoder :-

- Reconstructs the latent output to resemble the input.

3. Loss Function :- $L(x - \hat{x}) = ||(x - \hat{x})||^2$

4. Types :-

• Undercomplete Autoencoder :-

- Size of latent vector smaller than the input.

• Overcomplete Autoencoder :-

- Size of the latent vector greater than or equal to the input.

5. PCA i.e. Principle Component Analysis is another dimensionality reduction algorithm like Autoencoders.

6. We know that the loss function of autoencoders slightly resembles the Mean Squared Error formula.

7. Thus, an undercomplete encoder trained with respect to MSE learns a subspace equivalent to PCA.

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Q.26

8. In overcomplete autoencoders, the latent vector exceeds the input size i.e. $\geq$ input.

9. When regularisation is ^{not} done on such a vector, the extra exceeded size isn't normalised.

10. This results in making the size of the latent vector ~~almost equal~~ ^{much greater} to the original input size.

11. Thus, this makes the vector either copy the features or filled with noise and outliers.

12. In such cases, autoencoders tend to capture all the original identity features instead of only the essential ones, as the ~~original~~ ^{overcomplete} latent size is so large.

13. Thus ^{overcomplete} autoencoders should be regularised to remove noise and capture only essential features.

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~~Q.2a~~
Q.2a

Computational complexity of a standard 2d convolution.

Let

Input size (i) = 6×6

Kernel size (K) = 3×3

stride (s) = 1

Padding (p) = 0 [No padding]

Thus,

$$\text{Feature Map} = \left(\frac{i - K + 2p}{s} \right) + 1$$

$$= \left(\frac{6 - 3 + 2(0)}{1} \right) + 1$$

$$= 3 + 1$$

$$= 4$$

Thus, output image feature map would be of dimension 4×4 .

Hence, computational complexity will be $4 \times 4 = 16$

i.e. 16 multiplication operations would be required.

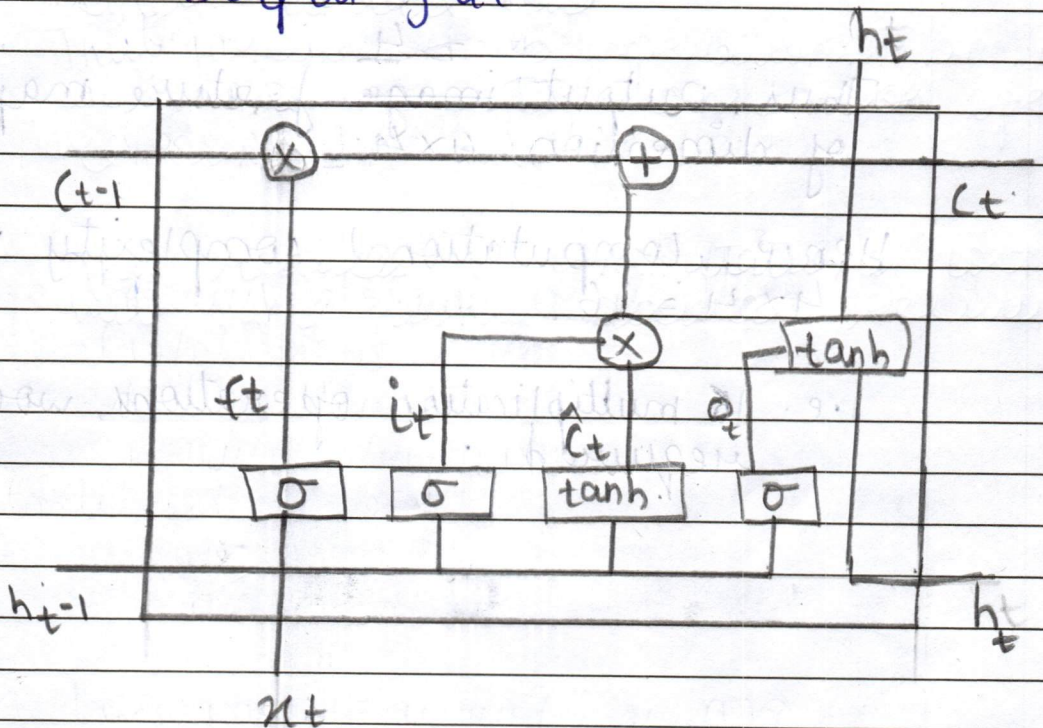
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Q.3a

1. The main disadvantage of a recurrent neural network is that as soon as the input pass goes on, the patterns or the context captured in the initial stage start to vanish.
2. This is called the famous vanishing gradient problem.
3. This is solved by a new type of model called Long-Short Term Memory (LSTM).
4. LSTM works on three different gates.
 - Input gate
 - Forget gate
 - Output gate.



LSTM.

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Q3a)

5. The ~~context~~^{output} coming from the previous cell stage is combined with the current state input and a sigmoid function is carried out between them.

$$f_t = \sigma[W_t(h_{t-1}, x_t) + b_t]$$

6. Thus ~~if~~ ~~this~~ this gives a new vector having 0's and 1's. 0's meaning the context has to be forgotten and 1 means the context has to be retained.

7. This new vector (f_t) is combined with the previous cell context and a pointwise operation is performed.

8. Pointwise operation is nothing but a simple one-to-one dot product.

9. Thus, if the new vector (f_t) has a '0' ~~the~~ the forget gate knows that the corresponding previous stage context has to be forgotten.

10. This goes on recursively and the gradient expression changes according to the changing context.

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Q3b) • LSTM has three gates

- Input gate
- Forget gate
- Output gate

• However, GRU (Gated Recurrent Unit) has only two gates

- Update Gate
- Reset Gate

~~LSTM:~~

- ~~Since~~ &

• GRU combines the operations done by the input and forget gate into one single update gate

• Thus, where 4-5 operations/junctions are done in - LSTM, only two operations are done in GRU.

• This reduces its computational cost with same highly accurate results.

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